



### 7.H2.6 Wrap-Up: Cool Down

1. Write a summary of what you learned today.

## Unit 7, Lesson H3: Area of a Triangle Using the Sine Ratio



### Benchmark

MA.912.T.1.4 Solve mathematical problems involving finding the area of a triangle given two sides and the included angle.



### Learning Target

- ☐ I can find the area of a triangle using the sine ratio.



### 7.H3.1 Warm-Up: History of Mathematics

Hannah Fry is an English mathematician who works alongside physicists, mathematicians, computer scientists, architects, and geographers to study the patterns in human behavior. Fry's research applies to a wide range of social problems and questions, from shopping and transport to urban crime, riots, and terrorism.

Fry has also written two popular math books: *The Mathematics of Love: Patterns, Proofs, and the Search for the Ultimate Equation* and *The Indisputable Existence of Santa Claus*.

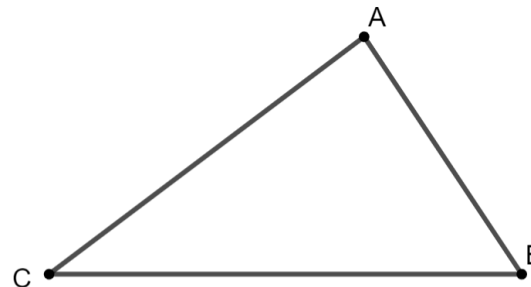
Fry is quoted as saying, "There is a mathematical angle to almost anything."

1. Interpret this quote in your own words.
2. What is a specific real-world example where mathematics is present?



### 7.H3.2 Exploration: Area of a Triangle

1.  $\triangle ABC$  is shown.



- a. Use  $\triangle ABC$  to do the following.
  - Draw an altitude from point  $A$  to side  $\overline{CB}$
  - Label the altitude as  $h$ .
  - Label the point where the altitude intersects side  $\overline{BC}$  as  $D$ .
  - Label the sides as  $AB = c$ ,  $BC = a$ , and  $AC = b$ .

The altitude creates two right triangles inside  $\triangle ABC$ .

- b. Write the equation that can be used to find the area of  $\triangle ABC$ .
- c. Complete each equation using right triangle trigonometry.

| $\triangle ADB$ | $\triangle ADC$ |
|-----------------|-----------------|
| $\sin(B) =$     | $\sin(C) =$     |

d. Rewrite each equation in terms of  $h$ .

| $\triangle ADB$ | $\triangle ADC$ |
|-----------------|-----------------|
| $h =$           | $h =$           |

e. Substitute each expression into the formula,  
 $A = \frac{1}{2}ah$ .

| $\triangle ADB$          | $\triangle ADC$          |
|--------------------------|--------------------------|
| $A = \frac{1}{2}a$ _____ | $A = \frac{1}{2}a$ _____ |

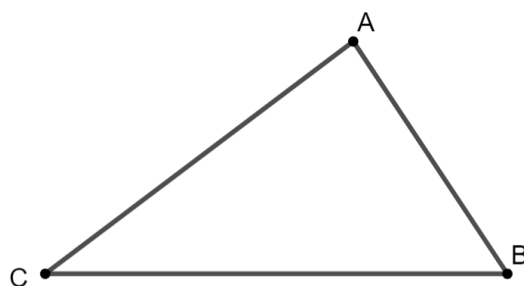
f. What do you notice about the two equations?

b. Find the area of the same triangle given that  
 $m\angle A = 90^\circ$ ,  $AC = 55$ , and  $AB = 48$ .

c. Discuss with your partner why the formula  $A = \frac{1}{2}bh$  could  
also be used when given  $m\angle A = 90^\circ$ ,  $AC = 55$ , and  $AB = 48$ .  
Summarize your discussion.

2.  $\triangle ABC$  is shown.

a. Find the area of  
 $\triangle ABC$  given that  
 $m\angle C = 41^\circ$ ,  
 $AC = 55$ , and  
 $BC = 73$ .





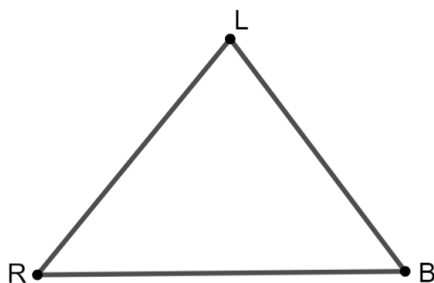
### 7.H3.3 Guided Practice: Area of a Triangle



The formulas for finding the area of a triangle using sine are:

$$A = \frac{1}{2}bc \sin A \quad A = \frac{1}{2}ac \sin B \quad A = \frac{1}{2}ab \sin C$$

1.  $\triangle RLB$  is shown.  $RL = 14$ ,  $RB = 16.5$ , and  $m\angle LRB = 50^\circ$ . Find the area of the triangle.



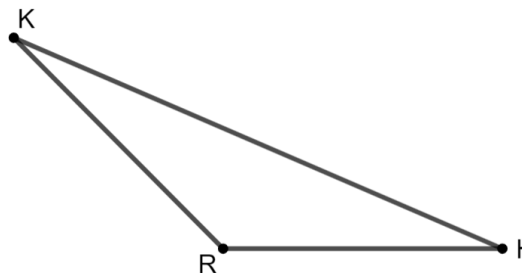
2. In  $\triangle FPB$ ,  $FP = 9$ ,  $PB = 18$ , and  $m\angle FPB = 117^\circ$ . Find the area of the triangle.



### 7.H3.4 Your Turn!

1. In  $\triangle NBR$ ,  $NB = 20$ ,  $NR = 18.6$ , and  $m\angle RNB = 43^\circ$ . Find the area of the triangle.

2.  $\triangle KRH$  is shown.  $KR = 18$ ,  $RH = 16$ , and  $m\angle KRH = 135^\circ$ . Find the area of the triangle.





### 7.H3.5 Wrap-Up: Cool Down

1. What are the similarities and differences between finding the area of a triangle and finding the area of a triangle using the sine ratio?

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